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**Tracking the Tide: An IoT-Based Smart Waste Sorting and Plastic Pollution
Monitoring System**

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Introduction

Today, plastic pollution stands as one of the most pervasive and rapidly worsening environmental challenges faced by communities around the world, affecting coastal cities, inland waterways, and even landlocked municipalities that feed waste into rivers leading to the sea. While plastic materials have become deeply embedded in modern life due to their affordability and convenience, the systems designed to manage plastic waste after use have failed to keep pace with the sheer scale of production. This mismatch between how quickly plastic is produced and how slowly it is properly collected, sorted, and recycled leaves individuals, waste management workers, and local governments without reliable, timely information on how much plastic waste is being generated, where it accumulates, and whether it is being diverted from landfills and waterways.

Throughout urban and coastal communities, residents and local officials have been struggling to cope with mounting volumes of plastic waste, often without adequate sorting infrastructure or real-time visibility into collection efficiency. In this sense, the absence of accurate, localized waste-tracking data results in continued leakage of plastic into rivers and oceans, contributing to an environmental and public health burden that continues to grow year after year. This problem of poorly monitored and inefficiently sorted plastic waste is a modern issue facing both developed and developing regions alike, and it is important to tackle by providing accessible technological solutions that empower individuals, waste collectors, and local governments with the data and tools they need to reduce plastic leakage. In this proposal, the author seeks to detail and scope the problem of plastic pollution and its effects on communities and ecosystems,

and to outline an ideal technology-enabled solution for smart waste sorting and pollution monitoring, creating an automated system to address the gap between plastic waste generation and effective waste management.

Problem description

In modern times, plastic pollution affects nearly every region of the world, not just coastal cities with direct ocean access, but also inland communities whose waste eventually reaches rivers and, ultimately, the sea. Global plastic production reached approximately 400 million tonnes per year by 2024, a dramatic increase from just 1.5 million tonnes in 1950, representing roughly a 260-fold rise in output over seven decades (Stop Ocean Plastic, 2026). Despite this explosive growth in production, waste management systems have not kept pace: of the roughly 8.3 billion tonnes of plastic produced since the 1950s, only 9 percent has ever been recycled, while 12 percent was incinerated and 79 percent accumulated in landfills or the natural environment (Stop Ocean Plastic, 2026). With this, the lasting impact of unmanaged plastic waste extends directly into aquatic ecosystems, as an estimated 8 to 11 million tons of plastic enter the ocean every year, equivalent to roughly 2,000 garbage trucks of plastic being dumped into oceans, rivers, and lakes every single day (Ocean Blue Project, 2026; Omni Calculator, 2026). Marine life bears a significant share of this burden, with around 100,000 marine mammals dying annually from plastic ingestion or entanglement, and over one in three fish caught for human consumption now found to contain microplastics (Omni Calculator, 2026).

With this observation in mind, it is worth noting how plastic pollution can affect both the environment and human health on a small and large scale, yet most households, waste

collectors, and local governments still lack a simple, reliable way of tracking how much plastic waste is generated, correctly sorted, and diverted from landfills or waterways. Some existing waste management efforts rely on manual sorting and infrequent, city-wide estimates, which are often slow, inconsistent, and unable to identify specific neighborhoods, barangays, or collection points where plastic leakage is most severe. Some challenges identified in the data are particularly relevant to communities located near rivers, informal dumpsites, or coastal areas, as generalized national or city-wide waste statistics often do not reflect the localized sorting and collection gaps these communities actually experience. With this in mind, the proposal of an ideal solution can be addressed by using smart sensor-based waste bins paired with a real-time monitoring and analytics platform to close the gap between plastic waste generation and effective, data-driven waste management.

Proposal solution

The proposed solution to address the lack of accurate, real-time plastic waste tracking is the development of an IoT-based Smart Waste Sorting and Plastic Pollution Monitoring System, referred to as EcoSort. This system is designed to help households, waste management personnel, and local government units systematically monitor, sort, and reduce plastic waste leakage through a network of smart collection bins and a connected mobile and web application. Sensor-equipped bins placed in residential areas, schools, markets, and along riverbanks use weight sensors, fill-level detectors, and basic image recognition to identify and log the type and volume of plastic waste deposited, automatically flagging bins that are full or that contain improperly sorted materials.

Using this data, the EcoSort mobile application displays real-time collection status and waste composition for each monitored location, enabling waste collectors to plan efficient pickup routes rather than relying on fixed schedules that often result in overflowing bins and street littering. One of the key features of the system is its automated pollution-risk alert function, which flags bins or collection points near waterways that are at high risk of overflow, notifying local sanitation offices to prioritize pickup before plastic waste can be washed into rivers or the ocean. Additional features include waste composition analytics, a community reporting tool for residents to flag illegal dumping or clogged waterways, monthly diversion-rate summaries for local governments, and gamified incentives that reward households and schools for consistent, correct plastic sorting.

The system primarily targets residents of urban and coastal communities, public school administrators, waste management personnel, and local government environmental offices, particularly in areas that currently lack dedicated waste-tracking infrastructure or that sit near rivers and coastlines vulnerable to plastic leakage. By automating waste detection, sorting verification, and collection prioritization, the system encourages more consistent recycling behavior, supports evidence-based local policymaking, and reduces the volume of plastic waste that ultimately reaches waterways and oceans. The expected impact of this solution includes increased plastic diversion and recycling rates, faster and more targeted waste collection near high-risk areas, reduced plastic leakage into rivers and coastlines, and a stronger evidence base for local governments to design targeted plastic-reduction interventions. Overall, integrating smart sensor-based waste bins with real-time monitoring and analytics offers

a sustainable, scalable, and community-driven approach to managing the growing crisis of plastic pollution.

Conclusion

Plastic pollution caused by rapidly growing production, low recycling rates, and inadequate waste-sorting infrastructure has become a persistent global issue affecting communities, marine ecosystems, and human health alike, leading to serious consequences such as ocean contamination, wildlife deaths, and widespread microplastic exposure. Addressing this problem requires not only awareness but also accessible, technology-enabled solutions that close the gap between plastic waste generation and effective, timely waste management. This proposal highlights the importance of implementing an IoT-based smart waste sorting and pollution monitoring system as a sustainable approach to managing plastic waste leakage at the community level. By tracking waste composition, prioritizing high-risk collection points, and encouraging consistent household sorting, the proposed system supports cleaner waterways and more responsive environmental policy. Ultimately, adopting automated, data-driven waste monitoring practices can help individuals and local institutions effectively respond to plastic pollution, protect marine and public health, and foster long-term environmental accountability.

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